IT3011 - Theory & Practices in Statistical Modelling

Hypothesis Testing

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Hypothesis Testing

When making a statistical inquiry, we often put forward a claim (hypothesis) about a population parameter.

Ex-

- Mean height of 15 year old boys is 1.62m
- The proportion of diabetes in a particular district is 0.05
- The mean number of telephone calls received by a telephone operator over a period of 60 minutes is 20.

The hypothesis we put forward is called the **Null Hypothesis** and is denoted by $\boldsymbol{H_0}$. Just define Null Hypotheses for above examples as below.

- $H_0 : \mu = 1.62$
- $H_0 : p = 0.05$
- $H_0 : \mu = 20$

In order to test the validity of H_0 we consider observations made on the random sample taken from the populations and perform a statistical test.

Hypothesis Testing

This test shows that whether we should reject the null hypothesis (rejecting the claim stated under H_0), we do so in favor of an **Alternative Hypothesis**, which is usually the opposite of what is stated in the null hypothesis and is denoted by $\boldsymbol{H_1}$ or $\boldsymbol{H_a}$.

Defining Hypotheses

When you get a question, an easy way to define hypotheses is, putting the question under the alternative hypothesis.

Ex-

1. The UGC Sri Lanka needs to know that whether the "majority of students at University of Moratuwa are from Western Province"

$$\boldsymbol{H_1 : p > 0.5} \quad \textit{and} \quad \boldsymbol{H_0 : p \leq 0.5} \longrightarrow \boldsymbol{H_0 : p = 0.5}$$

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.
3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.
4. A political news company believes the national approval rating for the current president has fallen below 40%.

Defining Hypotheses

2. A consumer test agency wants to see whether the mean lifetime of a brand of tires is less than 42000 miles as the tire manufacturer advertises that the average lifetime is at least 42000.

$$H_1 : \mu < 42000 \quad and \quad H_0 : \mu = 42000$$

3. The length of a certain lumber from a national home building store is supposed to be 8.5 feet. A builder wants to check whether the shipment of lumber she receives has a mean length different from 8.5 feet.

$$H_1 : \mu \neq 8.5 \quad and \quad H_0 : \mu = 8.5$$

4. A political news company believes the national approval rating for the current president has fallen below 40%.

$$H_1 : p < 0.4 \quad and \quad H_0 : p = 0.4$$

Type I Versus Type II Errors

When testing hypotheses there are essential two errors that can be made.

1. **Type I error** - Rejecting the null hypothesis when the null is correct actually.
2. **Type II error** - Not rejecting the null hypothesis when the null is not correct actually.

Decision	Reality	
	H_0 is true	H_0 is false
Reject H_0	Type I error	Correct
Do not reject H_0	Correct	Type II error

$$P(Type\ I\ error) = \alpha$$

$$P(Type\ II\ error) = \beta$$

Significance Level

When we conduct a hypothesis test we will always have to specify a significance level. We basically have to answer the question – “What is the probability of a Type I error we are willing to live with?”.

We cannot generally quantify the probability of a type II error. To avoid making Type ii errors we will always say “Do not reject the null hypothesis” instead of “Accept the null hypothesis”.

Popular Significance Levels

0.1 0.05 0.01